

The Culmination

A Formally-Verified Anatomical Substrate: Presenting the Governance Corpus as a Twelve-Organ Cybernetic Body with a Per-Theorem Verified Index

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Doctrine v11 LOCKED — 749 declarations · 14 unique axioms · 163 sorries @ c7c0ba17 · Λ is **Conjecture 1** · SLSA L1+L2 (not L3)

Abstract

This paper, v20 “The Culmination,” presents the SZL Holdings Ouroboros governance substrate as a single formally-verified *anatomical* object: a twelve-organ cybernetic body in which each organ is a typed runtime component, the organs communicate through an append-only, content-addressed event log, and the whole is held to a per-theorem verified index inherited from the v19 Verification Bridge. The contribution of v20 is not a new mathematical theorem; it is a *presentation* — the consolidation of the multi-track corpus into an anatomy that can be read, audited, and reasoned about as one body, with every claim carrying an explicit machine-checkable status. We organize the substrate’s proven core around the v18 thesis-theorem track (TH_V18_01–TH_V18_16), of which the majority are sorry-free over Lean-core axioms and a minority carry honestly-recorded open obligations, and around the five locked PURIQ governance formulas {F1, F11, F12, F18, F19}. We map each organ to the theorems that constrain it, the operational facts that establish it, and the open obligations that remain. Throughout we preserve the honesty doctrine verbatim: the Lutar invariant $\bigwedge_k(x) = (\prod_i x_i)^{1/k}$ is **Conjecture 1** unconditionally, never a theorem; exactly five formulas are locked; SLSA is L1+L2, not L3; every idealizing axiom is disclosed. v20 is the culmination of the expansion-then-consolidation arc: it is what the v18 expansion and the v19 bridge were *for* — a verified anatomy that the v21 runtime then executes and the v23 unification later closes.

Keywords: formal verification, Lean 4, cybernetic runtime, anatomical substrate, per-theorem index, honesty doctrine, governance algebra, event sourcing, Reed–Solomon, replay determinism.

*Honesty note (verbatim): The Lutar invariant Λ is **Conjecture 1** unconditionally and is never claimed proven unconditionally. Exactly five formulas are locked/proven at c7c0ba17. Modules that carry sorry obligations are reported as carrying them. Declared idealizations are disclosed. SLSA L1+L2 attested, not L3. Quechua organ names are brand naming, not prior-art or cultural claims; no mystical or physical-reality claims are made.*

1. What “culmination” means here

The Ouroboros thesis line is a sequence of versioned, DOI-pinned papers that record the provenance of the SZL governance substrate. v18 (“Multi-Track Substrate Expansion”) grew the Lean corpus across many tracks. v19 (“The Verification Bridge”) consolidated that corpus into a per-theorem verified index and imposed a strict locked-vs-experimental scope separation enforced by a CI drift gate. v20 is the **culmination** of that arc: it takes the verified index and *presents the whole substrate as one anatomical body*.

We are precise about the word. “Culmination” here does **not** mean “everything is now proven.” It means: the corpus has reached the point where it can be read as a single, legible, formally-grounded anatomy rather than as a pile of modules. The load-bearing prerequisite — that every claim carries a machine-checkable status — was delivered by v19. v20’s contribution is the organizing presentation that this status discipline makes possible.

Thesis of v20. A governance substrate is most trustworthy when it can be exhibited as an *anatomy*: a finite set of named organs, each with a typed interface, a deterministic step semantics, and an explicit ledger of which of its properties are kernel-verified, which are operational facts, and which are open. v20 constructs that anatomy over the existing corpus and shows that the verified-index discipline (v19) is exactly what makes the anatomy *verified* rather than merely *asserted*.

What v20 is not. v20 introduces no new proven mathematics. It closes no sorry that was not already closed at c7c0ba17. The conditional Λ -uniqueness theorem under a declared axiom, the machine-checked refutation of unconditional uniqueness, and the VCG truthfulness proof are all later (v22/v23) results and are *not* claimed here. v20’s honest scope is *presentation and organization*.

2. Related work: cybernetics, event sourcing, and verified runtimes

The anatomical framing is not metaphor for its own sake; it draws on three established traditions, each of which the substrate borrows from precisely and cites honestly.

2.1 Cybernetics and information theory

The organ-as-controller view is cybernetic in the original sense of [Wiener 1948]: control and communication in a system are realized through feedback loops over a shared channel. The substrate’s shared channel is the append-only event log, and the information carried on it is accounted for in the sense of [Shannon 1948]: each event is a content-hashed message, and the body’s observable state is exactly the sequence of such messages. We claim no mystical or biological reality for the anatomy; it is a disciplined engineering metaphor with a precise, event-sourced denotation.

2.2 Event sourcing as the anatomical substrate

That an entire runtime’s state can be reconstructed by replaying an append-only log of events is the event-sourcing pattern [Fowler 2005]. The substrate adopts it wholesale: inter-organ interactions are events on the Khipu log, so replay determinism (F1) and reciprocity conservation (F11) are statements about folds over that log, not about wall-clock behavior. This is what makes the anatomy *auditable*: the log is the body.

2.3 Aggregation theory and mechanism design

The aggregation organ (Ayni) and the Λ invariant sit in the theory of means and the economics of truthful mechanisms. The functional-equation characterizations of means [Aczél 1948] are the lineage in which the geometric mean is studied, and the row-geometric-mean ranking is characterized by a group-consensus axiom in [Csátó 2018] — precisely the block-consistency idea behind the declared axiom A6' under which later versions obtain a *conditional* uniqueness. The truthfulness target for the tool-call and voting organs draws on the Vickrey–Clarke–Groves tradition of dominant-strategy-incentive-compatible mechanisms; v20 carries only the framing, with the on-branch VCG truthfulness proof appearing in v22. We are careful that none of this is claimed proven in v20: it is the intellectual context that makes the Λ *conjecture* a well-posed open problem rather than a slogan.

2.4 Verified kernels and the per-theorem discipline

The verification posture is inherited from the verified-systems tradition — the seL4 OS kernel [Klein et al. 2009] and the Lean 4 prover and Mathlib [de Moura & Ullrich 2021], [mathlib Community 2020] — via the v19 bridge: a verified artifact states exactly what is proven and under what assumptions, and the anatomy reports each organ's status at the granularity of the theorem.

3. Formal preliminaries: organs, events, and the step semantics

Before presenting the body we fix the formal vocabulary the anatomy uses. The objects are organs and events; the semantics is a deterministic small-step relation over a shared log.

3.1 Organs, events, and the closed log

- **Event.** A content-hashed record (hop/role, payload, predecessor hash, self-hash) appended to the Khipu log. The log is append-only: no event is ever rewritten or deleted.
- **Organ.** A typed runtime component with a deterministic tick function that senses recent events and may emit new ones. An organ never mutates another organ's state directly; all interaction is mediated by the log.
- **Closed body.** The anatomy is *closed*: every inter-organ interaction is an event, so the body's observable state is exactly the log. This is the precondition that makes replay claims mechanical (§9).

3.2 The Lutar invariant and the candidate axioms

The aggregation organ (Ayni) is governed by the **Lutar invariant**, the equal-weight geometric mean

$$\Lambda_k(x_1, \dots, x_k) = (\prod_{i=1}^k x_i)^{1/k} \text{ on } [0,1]^k.$$

The candidate axioms a governed aggregator might satisfy — fixed here and carried verbatim into v22/v23 — are monotonicity (A1), positive homogeneity / ratio-scale invariance (A2), idempotence (A3), boundedness (A4), permutation invariance (A5), and the declared, governance-natural block-consistency axiom (A6'). v20 carries only the **label** on Λ (Conjecture 1) and the definitions; the conditional uniqueness theorem under A6' and the machine-checked refutation of unconditional uniqueness under {A1–A5} are later results.

4. The anatomical substrate: twelve organs

The substrate is presented as a twelve-organ cybernetic body, honestly cybernetic in the sense of Wiener and accounted for in the information-theoretic sense of Shannon: each organ emits content-hashed events into an append-only log, and another organ senses those events on its next tick. The twelve organs and their runtime roles:

Organ	Role in the anatomy
Puriq (core)	Scheduler/executor; ticks ready organs in fair round-robin.
Khipu	Append-only event log (content-addressed DAG segments).
Qillqaq	Serialiser/writer for Khipu records.
Unay	Receipt-keyed memory with semantic recall.
Ayni	Event-sourced reciprocity ledger.
Wayra	Always-learning ingest (chain-verified events).
Chaski	Reception/ingress edge organ (FIFO under backpressure).
Wallpa	Governed voice (open-source speech models only).
Wasi-Rikuq	Advisory observability (reports, never acts).
Hatun	MCP server exposing runtime tools.
Sentra	Mesh immune layer (anomaly detection + quarantine).
Hatun-Mind	Genome/DNA engine gating organ boot.

The anatomy is *closed*: every inter-organ interaction is an event on the Khipu log, so the body's observable state is exactly the log. This is what lets the substrate make replay claims mechanically rather than by assertion (§9).

4.1 The three subsystems of the body

The twelve organs group into three subsystems, each with a distinct accountability posture. Reading the body by subsystem clarifies why only a few organs carry locked theorems while most carry operational facts.

- **The verified core (Puriq, Khipu, Ayni).** These three sit on the verified spine and are backed by locked formulas: Puriq by replay determinism (F1) and agent-loop termination, Khipu by the Reed–Solomon and additive-entropy formulas (F18/F19) and the Kraft inequality, Ayni by reciprocity conservation (F11) and the $\wedge$ product form. They are the parts of the body whose correctness is *proved*, not merely observed.
- **The state and ingress organs (Qillqaq, Unay, Wayra, Chaski).** These handle serialization, memory, ingest, and edge reception. Their guarantees are largely *operational facts* — an LMDB write/kill/restart/read durability cycle for Unay, a 232-event chain-verified ingest for Wayra — with the corresponding formal obligations (F5/F6 recall and durability, F13 ingest verification) recorded as open.
- **The governance and effect organs (Wallpa, Wasi-Rikuq, Hatun, Sentra, Hatun-Mind).** These mediate voice, observation, tool calls, the immune layer, and boot gating. Their properties (OSS-only voice safety, advisory non-interference, tool idempotency, cross-cut completeness, genome totality) are mostly open obligations with declared discharge routes; the cross-organ agentic loop (§10) is where several of them gain a verified guarantee.

This grouping is itself an honesty device: it refuses to let the proved correctness of the core leak into an implicit claim that the whole body is proved. The verified core is small and named; everything outside it is tagged.

5. The verified spine: the v18 thesis-theorem track

The anatomy's *verified spine* is the v18 thesis-theorem track, sixteen modules under Lutar/Thesis/TH_V18_*, read verbatim from the corpus index. We report it with the v19 per-theorem honesty: sorry-free members are listed as such; members carrying open obligations are listed as carrying them.

Module (Lutar/Thesis/TH_V18_*)	sorry	ax	Status
TH_V18_01_AgentLoopTerminates	0	0	sorry-free
TH_V18_01_LambdaMonotonicity	0	0	sorry-free
TH_V18_02_DoctrineLabelFintype	0	0	sorry-free
TH_V18_03_KraftInequality	0	0	sorry-free
TH_V18_04_EgyptianWeightSum	0	0	sorry-free
TH_V18_05_ReceiptTransduction	1	0	open (1 obligation)
TH_V18_06_BrahmiAxisOption	1	0	open (1 obligation)
TH_V18_07_FeynmanCitationChain	1	0	open (1 obligation)
TH_V18_08_KhipuChecksumInvariant	3	0	open (3 obligations)
TH_V18_09_PermutationInvariance	0	0	sorry-free
TH_V18_10_ListSumInvariant	0	0	sorry-free
TH_V18_11_ParetoFiniteStabilization	1	0	open (1 obligation)
TH_V18_12_LambdaProductFormula	0	0	sorry-free
TH_V18_13_DPIBoundAbstract	0	0	sorry-free
TH_V18_14_SHA256CollisionHonest	0	2	sorry-free, axiom-gated
TH_V18_15_MultiAgentFairness	0	0	sorry-free
TH_V18_16_FeynmanCitationIntegrity	0	0	sorry-free

Eleven of the seventeen indexed thesis modules are sorry-free over Lean-core axioms; one (TH_V18_14) is sorry-free but *axiom-gated* — it declares two SHA-256 collision-resistance axioms and discloses them in the #print axioms ledger, exactly the disclosed-idealization discipline. Five carry honestly-recorded open obligations. This is the verified spine of the anatomy: it constrains agent-loop termination, the $\wedge$ product formula and its monotonicity/permutation invariance, the Kraft inequality, doctrine-label finiteness, the data-processing-inequality bound, and citation integrity.

Remark (the spine constrains the organs). The spine theorems are not decoration; they are the formal constraints the organs must respect. TH_V18_01_AgentLoopTerminates constrains the Puriq scheduler; TH_V18_09 and TH_V18_12 constrain the Ayni/aggregation semantics; TH_V18_03_KraftInequality constrains the Khipu/coding layer. The anatomy is ‘verified’ precisely in the sense that its spine is kernel-checked.

5.1 What the sorry-free spine theorems actually state

We expand the load-bearing sorry-free members so the spine is not just a count but a set of legible claims. Each is kernel-checked over Lean-core axioms only.

TH_V18_01 (Agent-loop termination) [sorry-free, Lean-core only]

The Puriq agent loop, modeled as a well-founded recursion on a decreasing work measure, terminates: there is no infinite tick sequence. This is the formal precondition for any liveness reasoning about the scheduler.

TH_V18_03 (Kraft inequality) [sorry-free, Lean-core only]

For any prefix-free code with codeword lengths ℓ_i , $\sum_i 2^{-\ell_i} \leq 1$. This constrains the Khipu coding layer: the content-addressed segment encoding cannot exceed the Kraft budget, so the log's addressing scheme is information-theoretically sound.

TH_V18_09 / TH_V18_12 (Permutation invariance & the $\wedge$ product form) [sorry-free, Lean-core only]

$\wedge_k$ is invariant under any reordering of its arguments (A5), and equals the k-th root of the product of its inputs. These two pin the Ayni aggregation semantics to the equal-weight geometric mean as a *definition*; they do **not** assert its uniqueness, which remains Conjecture 1.

TH_V18_13 (Data-processing-inequality bound, abstract) [sorry-free, Lean-core only]

Post-processing cannot increase the abstract information functional: $f(g(X))$ carries no more of the target signal than $g(X)$. This bounds what any downstream observability organ (Wasi-Rikuq) can recover, formalizing 'advisory observation does not amplify information.'

The remaining sorry-free members (TH_V18_02 doctrine-label finiteness, TH_V18_04 Egyptian weight-sum, TH_V18_10 list-sum invariance, TH_V18_15 multi-agent fairness, TH_V18_16 citation integrity) are structural lemmas that the organs rely on but that are individually elementary; we list them in the spine table above rather than expand each. The five members carrying open obligations (TH_V18_05–08, TH_V18_11) are reported with their exact sorry counts and are *not* claimed closed.

6. The five locked formulas, mapped onto the anatomy

The locked kernel at c7c0ba17 proves exactly five PURIQ governance formulas. v20 maps each onto the organ it governs, disclosing the #print axioms footprint of each.

- 1 **F1 — Replay determinism** (Puriq/Khipu). For a pure deterministic step f and input x , $f(x) = f(x)$, and over a recorded trace $xs.map\ f = xs.map\ f$. This is what makes the replay-hash gate admit every faithfully recorded step. #print axioms: *no axioms*.
- 2 **F11 — Ayni reciprocity conservation** (Ayni). $(b + c) - c = b$ over $\mathbb{Z}$: folding a credit then an equal debit conserves balance. This is fold-replay over an append-only ledger — not time travel. #print axioms: [propext].
- 3 **F12 — Kuramoto additive coupling** (Puriq scheduler phase term). $k \cdot (p_1 + p_2) = k \cdot p_1 + k \cdot p_2$ over $\mathbb{N}$. This is the *additive* scaffolding actually used; it is **not** the full nonlinear Kuramoto synchronization result. #print axioms: *no axioms*.

- 4 F18 — Reed–Solomon RS(10,6) recovery arithmetic** (Khipu durability). $10 - 6 = 4$ parity shards; erasing $e \leq 4$ shards leaves $10 - e \geq 6$ survivors, so the six data shards remain recoverable. This is integer bookkeeping over shard counts — **not** a holographic claim. #print axioms: [proptext, Quot.sound].
- 5 F19 — Bekenstein additive scaffolding** (Khipu entropy budget). $s_1 \leq s_1 + s_2$: the entropy budget of two disjoint regions is additive and monotone. The full Bekenstein bound $S \leq 2\pi kRE/(\hbar c)$ is **not** proved here; F19 is an explicit placeholder stub. #print axioms: *no axioms*.

F12 and F19 prove only *additive fragments* of, respectively, Kuramoto synchronization and the Bekenstein bound; they are honest scaffolding, never described as the full physical theorems.

7. Operational facts vs. proven theorems

The anatomy distinguishes three claim classes and never blurs them. v20 makes the distinction part of the anatomy itself: every organ property is tagged.

- **Proved.** Mechanised in Lean 4, sorry-free, depending only on Lean’s standard logical core (or a *disclosed* declared axiom). The verified spine (§5) and the five locked formulas (§6) are in this class.
- **Operational fact.** Established by running, signing, and verifying real artifacts — reproducible but not (yet) machine-proved. Examples carried into the anatomy from the runtime layer: a DSSE-signed artifact with a Rekor transparency-log entry, an LMDB write/kill/restart/read durability cycle, a chain-verified Wayra ingest.
- **Open.** Stated as such with a discharge route. The organ-level liveness, durability, and isolation properties of the runtime live here.

This three-class discipline is the same one v21 applies to its 23 agentic formulas (5 proved, 18 open). v20’s anatomy is the structure that makes those classifications legible across the whole body.

8. Organ-to-obligation traceability map

The value of presenting the substrate as an anatomy is that every organ becomes individually accountable: we can state, for each organ, which kernel-verified result constrains it, which operational fact establishes it, and which open obligation it still owes. The following traceability map is the v20 anatomy’s central audit artifact. It is honest about the asymmetry — only a few organs are backed by a locked theorem; most owe an open obligation with a declared discharge route.

Organ	Governing proved result	Open obligation owed	Claim class
Puriq (core)	F1 replay determinism; TH_V18_01 AgentLoopTerminates	F2 scheduler liveness	proved spine + open liveness
Khipu	F18 RS(10,6); F19 additive entropy; TH_V18_03 Kraft	F4 DAG acyclicity; F22 emit monotonicity	proved coding/entropy + open structure
Qillqaq	—	F22 single-writer append-only	open (operational single-writer)
Unay	—	F5 recall correctness; F6 LMDB durability	operational fact (write/restart cycle)
Ayni	F11 reciprocity; TH_V18_09/12 ($\wedge$ form)	F23 $\wedge$ -aggregator soundness (Conjecture 1)	proved reciprocity + open Conjecture 1

Organ	Governing proved result	Open obligation owed	Claim class
Wayra	—	F13 ingest chain-verification (232 events)	operational fact (chain verified)
Chaski	—	F7 FIFO under backpressure	open
Wallpa	—	F8 OSS-only voice safety	open (provenance predicate)
Wasi-Rikuq	—	F9 advisory non-interference	open (read-only effect typing)
Hatun	—	F10 tool-call idempotency	operational fact (16-tool MCP) + open
Sentra	—	F16 cross-cut completeness	open (defensive, advisory)
Hatun-Mind	—	F3 boot gating; F21 genome totality	open (validator totality)

The map makes the anatomy's honesty mechanical: a reader can point at any organ and read off exactly what is proven about it and what is still owed. Two organs — Puriq and Ayni — sit on the verified spine; Khipu is backed by two locked formulas; every other organ rests on operational facts and open obligations. Crucially, the Ayni row carries **Conjecture 1**: the Λ -aggregator soundness statement (F23) is the open obligation of the reciprocity/aggregation organ, and it is never rounded up to a theorem. This is the anatomy refusing to overclaim, organ by organ.

Remark (why a traceability map and not a proof of the whole body). A dishonest presentation would assert that the 'verified anatomy' is verified end-to-end. v20 does not. The map shows that most organs owe open obligations; the contribution is that those obligations are *named, located in a specific organ, and given a discharge route*, rather than hidden behind a blanket claim of verification. This is the v19 per-theorem discipline applied at the granularity of the runtime body.

9. Auditing the body: replay, the drift gate, and per-organ acceptance

An anatomy is only as trustworthy as the procedure that checks it. v20 inherits two complementary audit mechanisms from the v19 bridge and applies them at the granularity of the organ: *replay*, which re-derives the body's state from the log and compares hashes, and the *drift gate*, which refuses any change that would silently alter the locked counts. Together they convert the anatomy from a static diagram into a continuously re-checkable object.

9.1 Replay as the body's ground truth

Because the body is closed (§3), its entire observable state is the Khipu log, and any organ's behaviour can be re-derived by folding the relevant events. Replay determinism (F1) lifts this from a hope to a checkable property: replaying a recorded trace through the same deterministic tick functions must reproduce a byte-identical receipt chain. The audit procedure is therefore mechanical — recompute the fold, compare the terminal content-hash — and a mismatch localizes to the first divergent event, naming the organ that emitted it. This is why the anatomy can claim '5× byte-identical replay' as an operational fact rather than an aspiration.

Replay-audit invariant. For a recorded trace xs and the deterministic tick relation step, $\text{replay}(xs)$ yields a receipt chain whose self-hashes equal those recorded in xs . Any divergence is detected at the first event whose recomputed hash differs, so tampering or nondeterminism is *localizable* to a single organ emission. (Mechanizes F1 lifted to whole traces; #print axioms of the underlying F1: no axioms.)

9.2 The drift gate as the anatomy's acceptance test

The drift gate is a CI guard that recomputes the locked corpus statistics on every change and fails the build if they move without an explicit, reviewed baseline update. It is the operational embodiment of the 'locked = 5' doctrine: the gate makes it impossible to add experimental theorems into the locked namespace, or to silently introduce an axiom or a sorry, without the numbers changing and the gate tripping.

Drift-gate check	What it pins	Failure semantics
declaration count	749 locked declarations @ c7c0ba17	build fails on unreviewed delta
axiom count	14 unique declared axioms	build fails if a new axiom appears
sorry count	163 sorries (112 baseline + 51 Putnam)	build fails on hidden sorry change
locked-formula set	{F1, F11, F12, F18, F19}	build fails if the set changes
namespace separation	experimental theorems excluded from baseline	build fails if experimental leaks into locked

The gate is what lets v20 add a verified cross-organ pathway (§10) and cite experimental wave results *without* changing the locked numbers: the experimental work lives in a separate namespace the gate keeps out of the baseline. The anatomy can therefore grow new verified guarantees while the doctrine's 'exactly five locked' claim stays mechanically true.

Remark (acceptance is per-organ). Because every organ's status is tagged (§7–8), the acceptance test reads organ by organ: a change to Puriq must preserve F1 and TH_V18_O1; a change to Khipu must preserve F18/F19 and the Kraft bound; a change to Ayni must preserve F11 and must *not* promote the $\wedge$ aggregator beyond Conjecture 1. The drift gate enforces these per-organ invariants globally.

10. The agentic loop as a cross-organ pathway, verified end-to-end

The most consequential way the organs cooperate is the agentic control loop: retrieve (Chaski/Wayra) → plan (Puriq) → tool-call (Hatun) → policy-check (Sentra) → kernel-check (Hatun-Mind/doctrine) → emit (Wallpa/Khipu). Because the anatomy is closed, this pathway is itself an event sequence on the Khipu log, which means it can be modeled and verified as a system. An experimental-scope Lean development does exactly this, proving six end-to-end properties over whole runs of the loop — the anatomy's strongest cross-organ guarantee to date.

Property	Cross-organ guarantee	Status
P1 receipt-completeness	every hop leaves exactly one chained receipt on Khipu; a full loop $\Rightarrow$ 6 receipts	sorry-free
P2 gate-soundness	Wallpa emits iff both Sentra (policy) and the doctrine kernel ALLOW; one DENY is absorbing	sorry-free
P3 non-interference	adversary-controlled retrieval cannot flip a DENY to an ALLOW (Goguen-Meseguer)	axiom-free
P4 replay-determinism	replaying a loop from the same state reproduces a byte-identical receipt chain (F1 lifted)	axiom-free
P5 tamper-evidence	any single-receipt mutation is rejected on chain re-verification	axiom-gated (hash)
P6 monotone auditability	a Wasi-Rikuq auditor never retracts acceptance of a valid prefix as the log grows	sorry-free

Twenty-eight theorems carry these six properties (sorry-free, exit 0, no sorryAx); fourteen are fully axiom-free, ten use only the Lean-core trio, and four are axiom-gated on a single disclosed collision-resistance idealization [Goguen & Meseguer 1982]; [Merkle 1987]. The non-interference result — that a poisoned retrieval payload provably cannot flip a verdict — is the anatomy’s formal answer to prompt injection, and its core is axiom-free.

Scope note (honest). These 28 theorems are *experimental scope* in a separate namespace, NOT imported into the locked library and excluded from the drift baseline; the locked counts 749/14/163 at c7c0ba17 and locked_proven = 5 are UNCHANGED. The anatomy gains a verified cross-organ pathway without changing what the locked kernel proves.

11. The honesty doctrine, carried into the anatomy

v20 carries the v14 honesty doctrine forward unchanged and embeds it in the anatomy’s presentation. We restate it because every later version inherits it verbatim.

- 1** Λ is **Conjecture 1 unconditionally**. The Lutar invariant $\Lambda_k(x)$ is never claimed the unique aggregator under its axioms without a further declared assumption. (The machine-checked refutation of the unconditional claim and the conditional theorem under a declared block-consistency axiom are later v22/v23 results; v20 preserves the ‘ Λ = Conjecture 1’ label.)
- 2** **Locked = 5**. The locked kernel proves exactly {F1, F11, F12, F18, F19}. All other formal work is experimental until re-audited under the authoritative lake build.
- 3** **Disclosed idealizations**. Where a proof needs cryptographic hardness it declares a named axiom (the SHA-256 axioms in TH_v18_14) and discloses it; it does not pretend to prove hardness.
- 4** **SLSA L1+L2, not L3**.
- 5** **Open is open**. A sorry is reported as a sorry; a fragment as a fragment.

Conjecture 1 (The Lutar invariant; never a theorem). The equal-weight geometric mean Λ_k is the correct unique trust aggregator for governed AI. This is an open claim about the *right* axiomatization, not a mathematical theorem; the substrate carries the ‘Conjecture 1’ label on Λ in every artifact, including this one.

12. Empirical operating characteristics (measured, not proven)

The anatomy is cheap to operate; we report measured operating characteristics from the running body, each labeled **measured-not-proven**. These are reproducible operational facts about the organs, never theorems.

Quantity (organ)	Measured value	Conditions
Receipt build, Khipu (p50 / p99)	11.5 μ s / 50.7 μ s	content-hashed receipt emission
Receipt verify, Khipu	10.4 μ s	hash-chain link check
Λ_g aggregate, Ayni	3.12 / 3.29 μ s	9-axis geometric-mean evaluation
Governance overhead, Sentra (p50 / p99)	0.49–0.59 ms / 1.27 ms	over 24,800 governed calls
p-closure, Wasi-Rikuq	100% (8,000 / 8,000)	audit-closure pass rate
Replay determinism, Puriq	5 $\times$ byte-identical	repeated replay of a recorded trace
Wayra ingest	232 events chain-verified	always-learning ingest pathway

These figures characterize the running anatomy; they do not bear on what is proven. The honesty doctrine keeps the measured column strictly separate from the proven column — a measured latency is never described as a theorem, and a theorem is never described as a benchmark.

13. Position in the lineage

Ver	Date	Role
v18	2026-05-30	Expansion. 29 modules across many tracks; per-theorem index begun; open obligations recorded.
v19	2026-05-31	The Verification Bridge. Consolidation into a per-theorem verified index; locked-vs-experimental scope separation; the drift gate; honesty doctrine as acceptance criterion.
v20	2026-06-01	The Culmination (this paper). The verified anatomy: the substrate presented as a twelve-organ cybernetic body with a verified spine and a tagged claim ledger.
v21	2026-06-01	PURIQ-OS Substrate. The runtime that executes the verified anatomy; 23 agentic formulas, 5 proved.
v22	2026-06-03	Convergence. A5 structure-field merge, VCG proven on-branch, partial Cauchy closure, SLSA L2.
v23	2026-06-06	Unified Substrate. Conditional Λ -uniqueness under declared A6'; unconditional uniqueness machine-checked FALSE; Λ stays Conjecture 1.

The line reads as an unbroken lineage: v18 expands, v19 consolidates and verifies, **v20 presents the verified anatomy**, v21 ships the runtime, v22 converges, v23 unifies. v20 is the culmination of the expansion-then-consolidation arc and the substrate the runtime executes.

14. Limitations

We state plainly what v20 does *not* establish.

- 1 No new proven mathematics.** v20 closes no sorry and proves no new theorem; its contribution is the anatomical presentation and the claim-class tagging.
- 2 Open obligations remain.** Five thesis-spine modules carry open obligations (§5); most of the runtime organ properties are open formulas (§7).

- 3 Λ remains **Conjecture 1**. v20 preserves the label; it neither proves nor refutes uniqueness.
- 4 **Operational facts are not proofs**. DSSE/Rekor/LMDB/Wayra claims are reproducible operational facts, not machine-proved theorems.
- 5 **Physics analogies are additive scaffolding only**. F12 (Kuramoto) and F19 (Bekenstein) prove only additive fragments.
- 6 **SLSA L1+L2, not L3**.
- 7 **Counts are corpus-index counts**. Per-file counts are read from `lean_per_version.json`; the authoritative numbers are the locked kernel's (749/14/163) at `c7c0ba17`.

15. Open problems and discharge routes

The honest counterpart to the traceability map (§8) is a list of the open obligations it names, each with the route by which a future version intends to discharge it. v20 does not close these; it locates them and states how they could be closed, so the anatomy's debt is explicit rather than implicit.

Open obligation (organ)	Discharge route	Where pursued
F23 Λ-aggregator soundness (Ayni)	Conditional uniqueness under a declared block-consistency axiom A6'; unconditional uniqueness is machine-checked FALSE	v22 / v23
F2 scheduler liveness (Puriq)	Well-founded fairness measure extending TH_V18_01 termination to bounded-wait liveness	open
F6 LMDB durability (Unay)	Crash-consistency model over the write/kill/restart/read cycle; currently an operational fact	open
F9 advisory non-interference (Wasi-Rikuq)	Read-only effect typing + the agentic-loop P3 non-interference result (axiom-free core)	v20 §10 (partial)
F13 ingest verification (Wayra)	Lift the 232-event chain-verification operational fact to a fold invariant over the ingest pathway	open
F8 OSS-only voice safety (Wallpa)	A provenance predicate on model artifacts checked at boot by Hatun-Mind	open
F21 genome totality (Hatun-Mind)	Totality proof for the validator gating organ boot	open

Three observations make this list honest rather than aspirational. First, the Λ obligation (F23) is the deepest and is explicitly **Conjecture 1**: v22/v23 prove only a *conditional* uniqueness under a declared axiom and machine-check that the *unconditional* claim is false, so the obligation is reshaped, not removed. Second, several runtime obligations (F6, F13) already have a reproducible *operational fact* behind them; the open work is to lift the fact to a machine-checked invariant, not to discover whether the behaviour holds. Third, F9 already has a partial discharge via the agentic-loop non-interference result of §10, whose core is axiom-free. The map of debts is thus precise about which obligations are near and which, like Conjecture 1, are genuinely open research.

16. Conclusion

The Culmination is the version in which the governance corpus finally reads as a *body*. v20 presents the substrate as a twelve-organ cybernetic anatomy whose spine is the kernel-verified v18 thesis-theorem track and whose five locked formulas govern the organs that need them, with every other property honestly tagged as an operational fact or an open obligation. The anatomy is *verified* precisely because

the v19 bridge made per-theorem status auditable; without that discipline, the anatomy would be asserted rather than checked.

One-line thesis. v20 is the verified anatomy: the substrate presented as a twelve-organ cybernetic body whose spine is kernel-checked, whose five locked formulas govern their organs, and whose every remaining property is honestly tagged proved, operational, or open.

17. Appendix A: Reproducibility and pins

The artifacts behind every claim, pinned for reproduction.

Artifact	Pin / identifier
Locked kernel commit	lutar-lean @ c7c0ba17
Locked counts	749 declarations / 14 unique axioms / 163 sorries (112 baseline + 51 Putnam)
Locked formulas	{F1, F11, F12, F18, F19}
Verified spine	Lutar/Thesis/TH_V18_01 .. TH_V18_16 (corpus index)
Lean toolchain	Lean 4.13.0 (commit 6d22e0e5cc5a)
Concept DOI (always–latest)	10.5281/zenodo.19944926
v11 metrics DOI	10.5281/zenodo.20119582
ORCID	0009-0001-0110-4173

18. Appendix B: Organ glossary and claim–class legend

The twelve organs carry Quechua brand names (naming only; no prior-art or cultural claim). The claim classes are **proved** (kernel-checked, sorry-free, axioms disclosed), **operational fact** (reproducible but not machine-proved), and **open** (a named obligation with a discharge route).

Term	Meaning
closed body	Every inter-organ interaction is an event on the Khipu log; the body's state is the log.
verified spine	The kernel-checked TH_V18_* thesis-theorem track that constrains the organs.
operational fact	Established by running/signing/verifying real artifacts; reproducible, not yet proved.
Conjecture 1	The open claim that A6'-type block-consistency forces $\wedge$; never a theorem.
additive scaffolding	F12/F19 prove only additive fragments of Kuramoto/Bekenstein, never the full physics.
#print axioms	Lean command listing the trusted axiom base a theorem depends on.

19. References

- [1] Aczél, J. (1948). On mean values. *Bull. Amer. Math. Soc.* 54(4):392–400. doi:10.1090/S0002-9904-1948-09020-9
- [2] Bekenstein, J. D. (1981). Universal upper bound on the entropy-to-energy ratio for bounded systems. *Phys. Rev. D* 23(2):287–298. doi:10.1103/PhysRevD.23.287
- [3] Brundage, M., Avin, S., Wang, J., et al. (2020). Toward Trustworthy AI Development: Mechanisms for Supporting Verifiable Claims. arXiv:2004.07213. arxiv.org/abs/2004.07213
- [4] Csátó, L. (2018). Characterization of the row geometric mean ranking with a group consensus axiom. *Group Decision and Negotiation* 27(6):1011–1027. doi:10.1007/s10726-018-9589-3
- [5] de Moura, L. & Ullrich, S. (2021). The Lean 4 Theorem Prover and Programming Language. *CADE 28*, LNCS 12699, 625–635. doi:10.1007/978-3-030-79876-5_37
- [6] Fowler, M. (2005). Event Sourcing. [martinfowler.com](https://martinfowler.com/eaaDev/EventSourcing.html). martinfowler.com/eaaDev/EventSourcing.html
- [7] Goguen, J. A. & Meseguer, J. (1982). Security Policies and Security Models. *IEEE Symp. Security & Privacy*, 11–20. doi:10.1109/SP.1982.10014
- [8] Klein, G., Elphinstone, K., Heiser, G., et al. (2009). seL4: Formal Verification of an OS Kernel. *SOSP 2009*, 207–220. doi:10.1145/1629575.1629596

- [9] Kuramoto, Y. (1975). Self-entrainment of a population of coupled non-linear oscillators. *Int. Symp. Math. Problems in Theoretical Physics*, LNP 39, 420–422. doi:10.1007/BFb0013365
- [10] Lakatos, I. (1976). *Proofs and Refutations*. Cambridge University Press. [Proofs and Refutations](#)
- [11] The mathlib Community (2020). The Lean Mathematical Library. *CPP 2020*, 367–381. doi:10.1145/3372885.3373824
- [12] Merkle, R. C. (1987). A Digital Signature Based on a Conventional Encryption Function. *CRYPTO '87*, LNCS 293, 369–378. doi:10.1007/3-540-48184-2_32
- [13] Reed, I. S. & Solomon, G. (1960). Polynomial Codes Over Certain Finite Fields. *J. SIAM* 8(2):300–304. doi:10.1137/0108018
- [14] Shannon, C. E. (1948). A Mathematical Theory of Communication. *Bell Syst. Tech. J.* 27(3):379–423. doi:10.1002/j.1538-7305.1948.tb01338.x
- [15] Singleton, R. C. (1964). Maximum Distance q-ary Codes. *IEEE Trans. Inf. Theory* 10(2):116–118. doi:10.1109/TIT.1964.1053661
- [16] SLSA (2023). Supply-chain Levels for Software Artifacts, v1.0. OpenSSF. slsa.dev/spec/v1.0/levels
- [17] Wiener, N. (1948). *Cybernetics: Or Control and Communication in the Animal and the Machine*. MIT Press. [Cybernetics](#)

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Honesty doctrine preserved verbatim: Λ is Conjecture 1 unconditionally (never a theorem); the conditional uniqueness theorem holds only under the declared `A6'_block_consistent`; locked/proven = 5 {F1,F11,F12,F18,F19} @ c7c0ba17 (749/14/163); declared axioms disclosed in every `#print axioms ledger`; SLSA L1+L2 (not L3). No fabricated results, no fake citations. Quechua organ names are brand naming only.